

Beyond Average Return: Federated Reinforcement Learning via Adaptive Minimax Optimization

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Abstract—Federated reinforcement learning (FRL) aims to learn a shared policy across distributed environments without centralized data collection. Existing methods optimize an average-return objective, effectively solving a virtual MDP with averaged dynamics, which provides no per-environment performance guarantees under heterogeneity. To address this limitation, we propose a robust FRL framework based on a minimax objective over environments. A key challenge is that the worst-case environment is *policy-dependent* and evolves during training, rendering static robust formulations inadequate. We reformulate the problem as a saddle-point optimization and develop a primal–dual policy gradient method, termed *adaptive minimax policy optimization* (AMPO), which adaptively reweights environments to track those that limit worst-case performance. We establish sublinear convergence to a first-order stationary point and sublinear regret for the dual variable, implying that the policy is driven to improve performance on the most challenging environments, thereby promoting robustness across heterogeneous environments. Empirically, AMPO significantly improves worst-case performance while maintaining competitive average performance.

Index Terms—Federated reinforcement learning, robust optimization, saddle-point methods, environment heterogeneity.

I. INTRODUCTION

Learning a single policy that performs reliably across diverse environments is a central challenge in reinforcement learning (RL), particularly in distributed settings with heterogeneous dynamics. In such scenarios, heterogeneity in environment dynamics can lead to significant performance variability, making robust policy learning fundamentally challenging.

Federated reinforcement learning (FRL) has emerged as a promising paradigm to address these challenges by combining RL with federated learning (FL) [1]–[3]. In FRL, multiple agents interact with their own local environments and collaboratively learn a shared global policy through periodic aggregation of local updates, without sharing raw trajectories. This framework improves sample efficiency while preserving data privacy, making it particularly attractive for distributed systems.

A fundamental challenge in FRL is *environment heterogeneity*. Unlike standard RL, which optimizes a policy within a single environment, FRL requires learning a shared policy across multiple environments with distinct transition dynamics. This induces a fundamental trade-off: improving performance in one environment may degrade performance in others.

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Moreover, the state–action visitation distribution depends on the policy itself, making the heterogeneity inherently *policy-dependent*. This distinguishes FRL not only from standard RL, but also from conventional FL, where heterogeneity is fixed and arises solely from differences in data distributions, independent of the learned policy.

Most existing FRL approaches optimize an *average-return objective*, evaluating policies by averaging returns across environments. This formulation, inherited from empirical risk minimization in FL, is fundamentally misaligned with deployment: a policy is executed in a single environment, not an average of them.

More importantly, it effectively optimizes performance in a *virtual* environment induced by a mixture of local transition dynamics, which may not correspond to any real environment. As a result, policies learned under this objective may perform well for this virtual environment but offer no guarantees for any individual environment, and can fail in challenging regimes under heterogeneity.

These limitations highlight a fundamental mismatch between training and deployment, motivating the need for robust formulations that explicitly account for worst-case performance. To this end, we propose a robust FRL framework that adopts a minimax objective, aligning training with the requirement of reliable performance in each environment. Unlike prior approaches that rely on fixed aggregation rules, our method adaptively reweights environments, enabling policy updates that focus on performance-limiting regimes.

A. Related Works

Jin *et al.* [4] studied FRL under environment heterogeneity and proposed QAvg, a federated extension of Q-learning based on the standard averaging paradigm. Wang *et al.* [5] developed momentum-based policy gradient methods, including FedSVRPG-M and FedHAPG-M, to improve optimization efficiency via variance reduction and acceleration. Labbi *et al.* [6] established convergence guarantees for federated policy gradient methods under heterogeneous environments, achieving near-optimal stationary convergence with linear speedup. Zhu *et al.* [7] proposed a single-loop federated actor–critic (SFAC) algorithm that jointly updates actor and critic in a unified framework.

Despite these differences in algorithmic design and theoretical guarantees, existing FRL methods predominantly optimize an *average-return objective* across environments. This objective is fundamentally misaligned with deployment, where a policy is executed in a single environment and must perform

reliably in each case. Rather than ensuring strong performance across environments, it effectively optimizes performance in a *virtual* environment induced by a mixture of local transition dynamics, which can differ significantly from any individual environment. As a result, policies trained under this objective can perform poorly in individual environments, especially under strong heterogeneity.

This limitation exposes a fundamental gap: existing FRL methods do not explicitly account for worst-case performance. To address this, we propose a robust FRL framework that directly optimizes the worst-case return across environments. Importantly, unlike standard robust RL, where the worst-case scenario is defined over a fixed uncertainty set, the worst-case environment in FRL is inherently *policy-dependent* and evolves during training. Our framework enables adaptive tracking of such environments, rather than optimizing against a fixed worst-case scenario.

While our formulation shares a minimax structure with classical robust reinforcement learning, the underlying interpretation is fundamentally different. In robust RL, the worst-case is defined over a fixed uncertainty set of transition dynamics. In contrast, our setting considers a finite collection of heterogeneous environments, where the worst-case environment is inherently *policy-dependent* and evolves during training.

As a result, our framework captures a dynamic notion of robustness arising from environment heterogeneity, and enables adaptive tracking of performance-limiting environments rather than optimizing against a fixed adversarial scenario.

B. Contributions

We propose a robust FRL framework that departs from the standard average-return paradigm by adopting a minimax objective across heterogeneous environments. Our key insight is that robustness emerges from *adaptive reweighting* of environments, enabling the learning process to dynamically focus on performance-limiting regimes.

Our main contributions are summarized as follows:

- We introduce a minimax formulation for FRL that aligns the training objective with the deployment requirement of reliable per-environment performance, addressing the fundamental limitation of average-return objectives.
- We cast the problem as a saddle-point optimization via a primal–dual framework, where the dual variable dynamically reweights environments to track the policy-dependent worst-case environment. Based on this formulation, we develop an *adaptive minimax policy optimization* algorithm, termed AMPO.
- We establish convergence to a first-order stationary point of a dynamically reweighted objective, together with sub-linear regret guarantees for the dual variable. Our analysis shows that adaptive reweighting drives the policy toward robust performance across heterogeneous environments.
- We demonstrate that AMPO significantly improves worst-case performance while maintaining competitive average performance across heterogeneous environments.

II. FROM AVERAGE TO WORST-CASE: ROBUST FRL FORMULATION

We consider a federated reinforcement learning (FRL) setting with K environments (clients), each characterized by a distinct transition kernel P_k . The objective is to learn a single global policy π that generalizes across all environments. The discounted return of policy π under environment P_k is defined as

$$J_k(\pi) := \mathbb{E}_{P_k} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right], \quad (1)$$

where $\gamma \in (0, 1)$ denotes the discount factor.

Conventional FRL approaches optimize the average return across environments [4]:

$$\frac{1}{K} \sum_{k=1}^K J_k(\pi). \quad (2)$$

While this formulation is computationally convenient and consistent with standard federated learning, it does not adequately capture environment heterogeneity. In particular, the same policy induces different state-action visitation distributions across environments, and optimizing the average return effectively corresponds to maximizing performance in a *virtual environment* defined by a mixture of local transition dynamics. This virtual environment can differ significantly from any individual environment, making the objective less aligned with performance in any specific environment. As a result, this formulation may obscure poor performance in individual environments and provides no guarantee of satisfactory performance in each local environment.

Motivated by this limitation, we introduce a robust FRL formulation that explicitly accounts for *worst-case performance* across environments:

$$\max_{\pi} \min_{k \in [K]} J_k(\pi). \quad (3)$$

This formulation enforces robustness by directly optimizing the worst-case return, ensuring reliable performance across all environments. However, the worst-case environment is inherently *policy-dependent* and cannot be identified a priori, as the minimizing index may change as the policy evolves during learning. This induces a non-stationary optimization landscape and fundamentally distinguishes our setting from classical robust optimization, where the worst-case scenario is fixed.

To address this challenge, we adopt a primal–dual approach in which the dual variables define a distribution over environments. This reformulation converts the non-smooth worst-case objective into a tractable weighted objective, where the policy (primal variable) is optimized with respect to dynamically evolving environment weights. Meanwhile, the dual variables are updated to emphasize environments with lower returns, inducing an online reweighting process that adaptively tracks the policy-dependent worst-case environment. This perspective reveals an adaptive minimax optimization structure, in which learning continuously focuses on performance-limiting environments.

III. PRIMAL-DUAL REFORMULATION FOR ROBUST FRL

We present a primal-dual optimization framework for solving the robust FRL problem in (3). This reformulation provides a principled foundation for the adaptive minimax algorithm developed in subsequent sections.

A. Primal-Dual Formulation

To facilitate optimization, we introduce an auxiliary variable v and rewrite (3) in Hypograph form:

$$\begin{aligned} \max_{\pi, v} \quad & v \\ \text{s.t.} \quad & J_k(\pi) \geq v, \quad \forall k \in [K], \end{aligned} \quad (4)$$

where v represents the worst-case return across environments. Maximizing v therefore corresponds to improving worst-case performance.

We derive a tractable formulation via Lagrangian relaxation. The Lagrangian of (4) is given by

$$\mathcal{L}(\pi, v, \lambda) = v + \sum_{k=1}^K \lambda_k (J_k(\pi) - v), \quad \lambda_k \geq 0. \quad (5)$$

Rearranging terms yields

$$\mathcal{L}(\pi, v, \lambda) = \sum_{k=1}^K \lambda_k J_k(\pi) + \left(1 - \sum_{k=1}^K \lambda_k\right) v. \quad (6)$$

For the maximization over v to remain finite, we require

$$\sum_{k=1}^K \lambda_k = 1, \quad (7)$$

which implies that λ lies on the probability simplex

$$\Delta_K := \left\{ \lambda \in \mathbb{R}_+^K : \sum_{k=1}^K \lambda_k = 1 \right\}. \quad (8)$$

Substituting this condition, we obtain the following saddle-point formulation:

$$\min_{\lambda \in \Delta_K} \max_{\pi} \sum_{k=1}^K \lambda_k J_k(\pi). \quad (9)$$

Here, the dual variable λ defines a distribution over environments, and the policy π maximizes a weighted sum of returns. Importantly, this formulation enables adaptive emphasis on environments that limit performance.

Remark 1: In general, strong duality does not hold due to the non-convexity of policy optimization in reinforcement learning. Therefore, (9) can be interpreted as a relaxation of the original robust objective. Nevertheless, weak duality always holds:

$$\max_{\pi} \min_{k \in [K]} J_k(\pi) \leq \min_{\lambda \in \Delta_K} \max_{\pi} \sum_{k=1}^K \lambda_k J_k(\pi).$$

Strong duality can be recovered in structured settings, such as tabular discounted problems formulated over occupancy measures, which yield linear programs.

B. Primal-Dual Updates

Based on the saddle-point formulation in (9), we consider an alternating primal-dual optimization procedure.

Primal update (policy update). Given the current dual weights $\lambda^{(m)}$, the policy is updated to maximize the weighted objective:

$$\pi^{(m+1)} = \arg \max_{\pi} \sum_{k=1}^K \lambda_k^{(m)} J_k(\pi). \quad (10)$$

In practice, this maximization is approximated via policy gradient updates, whose specific form is detailed in the next section.

Dual update (environment reweighting). After evaluating the environment-wise returns $J_k(\pi^{(m+1)})$, the dual variables are updated via projected gradient descent to emphasize environments with lower returns:

$$\lambda^{(m+1)} = \Pi_{\Delta_K} \left(\lambda^{(m)} - \eta_{\lambda} \begin{bmatrix} J_1(\pi^{(m+1)}) \\ \vdots \\ J_K(\pi^{(m+1)}) \end{bmatrix} \right), \quad (11)$$

where Π_{Δ_K} denotes projection onto the simplex.

While (9) resembles a standard saddle-point problem, it admits a distinct interpretation in robust reinforcement learning. The worst-case environment is generally *policy-dependent*, meaning that the index attaining $\min_k J_k(\pi)$ may change as the policy evolves. As a result, the problem cannot be reduced to optimizing a fixed environment, but instead requires improving performance on environments that currently limit the objective.

From this viewpoint, the dual variable λ acts as an adaptive reweighting mechanism that assigns larger weights to environments with lower returns. This induces an online reweighting process over environments, enabling adaptive tracking of the policy-dependent worst-case environment. Correspondingly, the policy π is optimized with respect to a dynamically reweighted objective, focusing on improving performance in the environments emphasized by λ .

This interaction induces an *adaptive minimax optimization* process, in which the policy is repeatedly optimized under shifting emphasis on challenging environments, thereby improving worst-case performance.

IV. POLICY GRADIENT FOR ADAPTIVE MINIMAX FRL

We develop a policy gradient method to solve the primal update in (10) within the proposed primal-dual framework. The primal update requires maximizing a weighted sum of returns across heterogeneous environments:

$$\max_{\pi} J_{\lambda}(\pi) := \sum_{k=1}^K \lambda_k J_k(\pi), \quad (12)$$

where $\lambda \in \Delta_K$ is fixed.

A natural question is whether value-based methods such as Q-learning can be applied. However, even in conventional FRL formulations based on average-return objectives, a unified Bellman optimality equation does not naturally arise, as the

problem does not correspond to a single MDP. Although a mixture MDP interpretation is possible, the resulting Bellman equation is misaligned with the original FRL objective. This issue is further exacerbated in our setting, where the worst-case objective induces a policy-dependent and non-stationary optimization problem, precluding a well-defined Bellman update.

In contrast, policy gradient methods are directly applicable, as they optimize the objective over trajectories without requiring a unified Bellman structure, making them well-suited for heterogeneous FRL settings with environment-dependent dynamics.

The weighted objective admits the following stochastic interpretation: at the beginning of each episode, an environment index k is sampled according to λ , and a trajectory τ is generated by executing policy π under dynamics P_k . Under this sampling process, we define

$$J_\lambda(\pi) = \mathbb{E}_{k \sim \lambda, \tau \sim (\pi, P_k)} [R(\tau)], \quad (13)$$

where $R(\tau) := \sum_{t=0}^{T-1} \gamma^t r_t$. The following theorem characterizes the policy gradient of the weighted objective.

Theorem 1 (Extended Policy Gradient Theorem): Let $\lambda \in \Delta_K$ be fixed, and let $\pi_\theta(a|s)$ be a differentiable stochastic policy. Then the gradient of $J_\lambda(\pi_\theta)$ is given by

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^K \lambda_k \mathbb{E}_{\tau \sim (\pi_\theta, P_k)} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) G_t^{(k)} \right], \quad (14)$$

where $G_t^{(k)}$ denotes the return-to-go at time t under environment k .

Theorem 1 shows that the gradient decomposes into a weighted sum of environment-wise policy gradients, aligning naturally with the FRL paradigm, where each client computes a local gradient and the server aggregates them according to the dual weights λ .

Importantly, the worst-performing environment is generally *policy-dependent*, meaning that the index attaining $\min_k J_k(\pi_\theta)$ may change as the policy evolves. Consequently, learning cannot be reduced to optimizing a fixed environment, but instead requires improving performance on environments that currently limit the objective. From this perspective, the dual variable assigns larger weights to environments with lower returns, inducing an adaptive reweighting of the policy gradient and enabling implicit online tracking of the worst-case environment.

This interaction induces an *adaptive minimax optimization* process, in which the policy is optimized under evolving environment weights, thereby promoting robust performance across heterogeneous environments.

V. AMPO: ADAPTIVE MINIMAX POLICY OPTIMIZATION FOR ROBUST FRL

We develop a federated reinforcement learning algorithm, termed *AMPO*, for solving the saddle-point formulation in (9). AMPO realizes an *adaptive minimax policy optimization* framework, in which the policy is optimized under dynamically evolving environment weights.

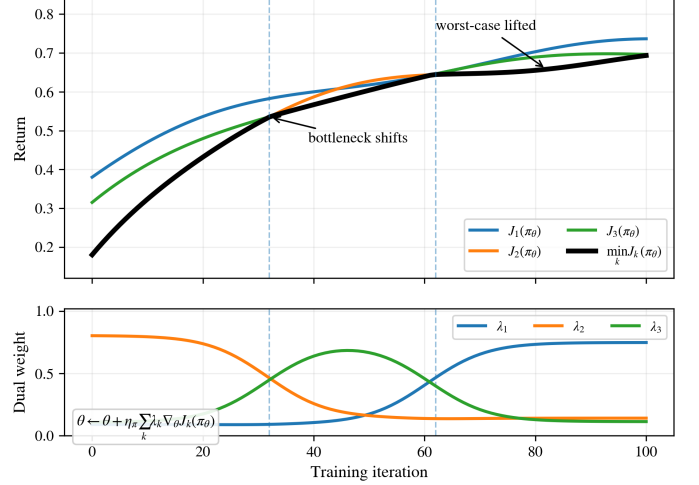


Fig. 1: Illustration of adaptive dual reweighting in the proposed framework. During training, the dual variables $\{\lambda_k\}$ emphasize the current bottleneck environment, guiding policy updates that improve its performance. As the worst-performing environment changes, the minimum return $\min_k J_k(\pi_\theta)$ increases, resulting in a balanced and robust policy across environments.

Building on the primal–dual formulation, AMPO alternates between policy optimization and environment reweighting. The policy is optimized with respect to a dynamically reweighted objective, while the dual variables are updated to assign larger weights to environments with lower returns. This enables adaptive tracking of the policy-dependent worst-case environment, and induces a learning process that focuses on performance-limiting environments, thereby improving the worst-case return across heterogeneous environments.

AMPO follows a client–server architecture. At each iteration, clients compute local policy gradients and return estimates using their respective environments, and the server aggregates these signals to update the global policy. In parallel, the server updates the dual variables to emphasize environments with lower returns.

A. The AMPO Algorithm

The proposed AMPO algorithm proceeds by alternating between local updates at the clients and global updates at the server, following a primal–dual optimization procedure.

Local Update (Clients). At iteration m , the server broadcasts the current global policy parameter $\theta^{(m)}$ to all clients. Each client k interacts with its local environment P_k using the policy $\pi_{\theta^{(m)}}$ to collect trajectories. Based on these trajectories, each client computes a sample-based estimate of the local policy gradient:

$$g_k^{(m)} \approx \mathbb{E}_{\tau \sim (\pi_{\theta^{(m)}}, P_k)} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_{\theta^{(m)}}(a_t|s_t) \hat{G}_t^{(k)} \right], \quad (15)$$

where $\hat{G}_t^{(k)}$ denotes an estimator of the return-to-go or advantage under environment P_k . Each client also estimates its local

return

$$J_k^{(m)} := \hat{J}_k(\pi_{\theta^{(m)}}), \quad (16)$$

via Monte Carlo rollouts, and transmits $(g_k^{(m)}, J_k^{(m)})$ to the server.

Global Update (Server). Upon receiving $\{g_k^{(m)}, J_k^{(m)}\}_{k=1}^K$, the server performs the following updates:

- Policy update:

$$g^{(m)} = \sum_{k=1}^K \lambda_k^{(m)} g_k^{(m)}, \quad \theta^{(m+1)} = \theta^{(m)} + \eta_\pi g^{(m)}. \quad (17)$$

This corresponds to gradient ascent on the dynamically reweighted objective.

- Dual update:

$$\lambda^{(m+1)} = \Pi_{\Delta_K} \left(\lambda^{(m)} - \eta_\lambda \begin{bmatrix} J_1^{(m)} \\ \vdots \\ J_K^{(m)} \end{bmatrix} \right). \quad (18)$$

This update increases the weights of environments with lower returns, encouraging improvement in the worst-performing environments.

The proposed algorithm admits a clear interpretation as an adaptive minimax optimization procedure. The dual variable λ upweights environments with lower returns, thereby steering the policy toward improving its worst-case performance. Importantly, λ is not a fixed aggregation coefficient or a manually tuned importance weight; it is an optimization variable derived from the minimax formulation. Therefore, the weighted gradient aggregation in AMPO should be viewed as the primal update of a saddle-point problem, rather than as a heuristic weighted averaging scheme.

B. Convergence and Regret Analysis

We analyze the convergence behavior of AMPO from the perspective of stochastic saddle-point optimization. Let

$$F(\theta, \lambda) := \sum_{k=1}^K \lambda_k J_k(\pi_\theta), \quad \lambda \in \Delta_K, \quad (19)$$

where π_θ denotes the policy parameterized by θ . The algorithm performs stochastic gradient ascent on θ and projected gradient descent on λ . To facilitate the analysis, we adopt the following standard assumptions. Let $\mathcal{H}_m$ denote the history up to iteration m .

Assumption 1: Each $J_k(\pi_\theta)$ is L -smooth in θ .

Assumption 2: The policy gradient estimator $g^{(m)}$ is conditionally unbiased with bounded variance:

$$\begin{aligned} \mathbb{E}[g^{(m)} \mid \mathcal{H}_m] &= \nabla_\theta F(\theta^{(m)}, \lambda^{(m)}), \\ \mathbb{E}[\|g^{(m)} - \nabla_\theta F(\theta^{(m)}, \lambda^{(m)})\|^2 \mid \mathcal{H}_m] &\leq \sigma^2. \end{aligned}$$

Assumption 3: The return estimator $\hat{J}_k^{(m)}$ is conditionally unbiased with bounded second moment:

$$\begin{aligned} \mathbb{E}[\hat{J}_k^{(m)} \mid \mathcal{H}_m] &= J_k(\pi_{\theta^{(m)}}), \\ \mathbb{E}[\|\hat{J}_k^{(m)} - J_k(\pi_{\theta^{(m)}})\|^2 \mid \mathcal{H}_m] &\leq \sigma_J^2. \end{aligned}$$

Assumption 4: The iterates $\{(\theta^{(m)}, \lambda^{(m)})\}$ remain in a bounded domain.

We now characterize the convergence behavior of the proposed primal–dual updates under the above assumptions.

Theorem 2 (Primal Convergence and Dual No-Regret): Under Assumptions 1)–4), with step sizes $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$, the iterates generated by AMPO satisfy

$$\begin{aligned} \frac{1}{M} \sum_{m=0}^{M-1} \mathbb{E}[\|\nabla_\theta F(\theta^{(m)}, \lambda^{(m)})\|^2] &= \mathcal{O}(M^{-1/3}), \\ \sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda^{(m)}) - \min_{\lambda \in \Delta_K} \sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda) &\leq \mathcal{O}(M^{2/3}). \end{aligned}$$

The slower regret rate compared to the classical $\mathcal{O}(\sqrt{M})$ arises from the coupling between the primal and dual variables, where the dual update operates on a sequence of policy-dependent objectives that evolve with the policy. Notably, when the environment weights are fixed, the problem reduces to standard policy optimization with a static objective, for which the classical $\mathcal{O}(1/\sqrt{M})$ convergence rate can be recovered. The slower rate in our result thus reflects the additional challenge of adaptive, policy-dependent reweighting.

The first result shows that the policy converges to a first-order stationary point of a *dynamically reweighted objective*. Although the objective is non-stationary due to the evolving dual weights, this ensures that the learning dynamics remain stable under adaptive reweighting. This is particularly important in our setting, where the objective evolves due to policy-dependent reweighting. The second result establishes that the dual updates achieve sublinear regret over the simplex, implying that the adaptive weighting $\{\lambda^{(m)}\}$ performs nearly as well as the best fixed weighting in hindsight. Consequently, the algorithm avoids persistently focusing on suboptimal environments and instead consistently emphasizes those that limit performance.

Taken together, these results provide a principled justification for AMPO. Despite the absence of a fixed worst-case environment and the resulting non-stationarity, the dual variable reliably tracks performance-limiting environments, while the policy adapts accordingly. This interaction continuously directs learning toward the most challenging regimes, thereby promoting robust performance across heterogeneous environments.

Finally, the guarantees are established in an ergodic sense, which is standard in stochastic and saddle-point optimization. While this does not imply optimality of the last iterate, it ensures that the time-averaged behavior of the algorithm approaches a robust solution in terms of worst-case performance across environments, and that the learning dynamics remain stable and well-directed throughout training.

Remark 2 (Practical choice of learning rates): The rates in Theorem 2 suggest a practical two-timescale update rule for AMPO. Since $\eta_\lambda = \mathcal{O}(M^{-2/3})$ decays faster than $\eta_\pi = \mathcal{O}(M^{-1/3})$, the dual variable should be updated more conservatively than the policy parameter. In practice, this motivates choosing $\eta_\lambda < \eta_\pi$. A large dual step size may cause the weights λ to react too aggressively to noisy return

estimates, leading to oscillatory worst-case tracking, whereas an overly small dual step size makes the method close to uniform aggregation. Thus, the theoretical step-size scaling supports using a moderately smaller dual learning rate to achieve stable adaptive reweighting.

C. Robustness–Scalability Trade-off

We analyze the impact of adaptive dual weighting on the scalability of federated policy optimization. In standard FRL methods based on the average-return objective, local gradients are uniformly averaged across K clients, leading to variance reduction proportional to $1/K$ and enabling linear speedup.

In contrast, AMPO aggregates gradients using adaptive weights:

$$\nabla_{\theta} F(\theta, \lambda) = \sum_{k=1}^K \lambda_k \nabla_{\theta} J_k(\pi_{\theta}), \quad (20)$$

where $\lambda \in \Delta_K$ evolves to emphasize environments with lower returns.

Proposition 1 (Effective Parallelism under Adaptive Weighting): Under the conditional independence assumption on local gradient estimators $\{g_k\}_{k=1}^K$ given $\mathcal{H}_m$, and assuming each estimator is unbiased with bounded variance σ^2 (as in Assumption 2), the variance of the aggregated gradient satisfies

$$\mathbb{E} \left[\left\| \sum_{k=1}^K \lambda_k (g_k - \nabla_{\theta} J_k) \right\|^2 \middle| \mathcal{H}_m \right] \leq \sigma^2 \sum_{k=1}^K \lambda_k^2. \quad (21)$$

This result motivates defining the *effective number of environments* as

$$K_{\text{eff}} := \frac{1}{\sum_{k=1}^K \lambda_k^2}, \quad (22)$$

which characterizes the effective degree of variance reduction in the aggregated gradient. When λ is uniform, i.e., $\lambda_k = \frac{1}{K}$, we recover $K_{\text{eff}} = K$, corresponding to classical linear speedup. However, as λ concentrates on a subset of environments, K_{eff} decreases, reducing the effective variance reduction.

This analysis reveals a fundamental trade-off: adaptive reweighting improves robustness by emphasizing challenging environments, but reduces effective parallelism. Consequently, AMPO achieves a *partial speedup* governed by K_{eff} , rather than full linear speedup in K . Importantly, this trade-off is intrinsic to adaptive minimax optimization: prioritizing difficult environments improves robustness but necessarily limits variance reduction. This provides a quantitative characterization of the robustness–scalability trade-off in FRL.

VI. PPO-BASED IMPLEMENTATION OF AMPO

In this section, we present a practical instantiation of the proposed AMPO framework using Proximal Policy Optimization (PPO) within an actor–critic architecture. This instantiation provides a stable and scalable implementation of AMPO in continuous control settings. While we adopt PPO as a stable backbone, the AMPO framework is general and can be seamlessly combined with other policy optimization methods, such as TRPO, A2C/A3C, and SAC.

Under AMPO, policy optimization is performed with respect to a dynamically reweighted objective, where environment importance is determined by the dual variables. PPO serves as the underlying policy optimization method, enabling stable updates under stochastic policy gradients.

We consider a shared stochastic policy $\pi_{\theta}(a | s)$ parameterized by θ , which is maintained by the server and broadcast to all clients. Each client k interacts with its local environment P_k and maintains an environment-specific critic V_{ψ_k} for value estimation. This design is well suited to federated reinforcement learning, where clients may have heterogeneous transition dynamics and reward landscapes. Unlike multi-agent RL methods such as MAPPO, which typically assume agents interacting within a shared environment and rely on a centralized critic, our setting consists of multiple heterogeneous environments. Thus, local critics naturally capture client-specific value functions while sharing a common robust policy.

Local Update (Clients). At iteration m , each client k receives the global policy parameter $\theta^{(m)}$ from the server and collects trajectories by interacting with its local environment P_k .

Each client updates its local critic V_{ψ_k} using temporal-difference (TD) learning. Based on the collected trajectories, the advantage is estimated using generalized advantage estimation (GAE):

$$A_t^{(k)} = \sum_{l=0}^{\infty} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}^{(k)}, \quad (23)$$

$$\delta_t^{(k)} = r_t + \gamma V_{\psi_k}(s_{t+1}) - V_{\psi_k}(s_t), \quad (24)$$

where $\lambda_{\text{GAE}} \in [0, 1]$ controls the bias–variance trade-off.

Using the estimated advantages, each client forms the PPO surrogate objective:

$$\mathcal{L}_k^{\text{PPO}}(\theta) = \mathbb{E} \left[\min \left(r_t(\theta) A_t^{(k)}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t^{(k)} \right) \right], \quad (25)$$

where

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta^{(m)}}(a_t | s_t)}. \quad (26)$$

Each client performs multiple PPO updates based on mini-batches of collected trajectories and computes a stochastic policy gradient estimate

$$g_k^{(m)} = \nabla_{\theta} \mathcal{L}_k^{\text{PPO}}(\theta) \big|_{\theta=\theta^{(m)}}. \quad (27)$$

In addition, each client estimates the local return

$$J_k^{(m)} := J_k(\pi_{\theta^{(m)}}), \quad (28)$$

via Monte Carlo rollouts. The pair $(g_k^{(m)}, J_k^{(m)})$ is then transmitted to the server.

Global Update (Server). Upon receiving $\{g_k^{(m)}, J_k^{(m)}\}_{k=1}^K$ from all clients, the server performs the policy and dual updates.

i) Policy update. The server aggregates local gradients using the current dual weights:

$$g^{(m)} = \sum_{k=1}^K \lambda_k^{(m)} g_k^{(m)}, \quad (29)$$

$$\theta^{(m+1)} = \theta^{(m)} + \eta_{\pi} g^{(m)}. \quad (30)$$

Algorithm 1 AMPO with PPO Backbone

Require: Initial actor parameter $\theta^{(0)}$, dual variable $\lambda^{(0)} \in \Delta_K$, number of rounds M

- 1: **for** $m = 0, 1, \dots, M - 1$ **do**
- 2: **Server broadcast:** send $\theta^{(m)}$ to all clients
- 3: **for each client k in parallel do**
- 4: Collect trajectories in environment P_k using $\pi_{\theta^{(m)}}$
- 5: Update local critic V_{ψ_k} via TD learning
- 6: Compute advantages $A_t^{(k)}$ using GAE
- 7: Perform multiple PPO updates and obtain a stochastic gradient estimate $g_k^{(m)}$
- 8: Estimate return $J_k^{(m)}$ via Monte Carlo rollouts
- 9: Send $(g_k^{(m)}, J_k^{(m)})$ to the server
- 10: **end for**
- 11: **Server update:**
- 12: Perform policy update via (29) and (30)
- 13: Perform dual update via (31)
- 14: Broadcast updated policy $\theta^{(m+1)}$ to all clients
- 15: **end for**

This corresponds to gradient ascent on a dynamically reweighted objective, where environments with larger dual weights contribute more to the update.

ii) *Dual update.* The server updates the dual variable based on the observed returns:

$$\lambda^{(m+1)} = \Pi_{\Delta_K} \left(\lambda^{(m)} - \eta_\lambda \begin{bmatrix} J_1^{(m)} \\ \vdots \\ J_K^{(m)} \end{bmatrix} \right), \quad (31)$$

where $\eta_\lambda > 0$ is the dual step size and Π_{Δ_K} denotes projection onto the probability simplex.

This update increases the weights of environments with lower returns, thereby steering the shared policy toward improving its worst-case performance. As a result, the policy is optimized under dynamically evolving environment weights, realizing the adaptive minimax optimization behavior of AMPO.

VII. EXPERIMENTS

A. Experimental Setup

B. Worst-Case Performance Analysis

C. Overall Performance under Heterogeneous Environments

D. Dual Variable Dynamics

E. Ablation Study

APPENDIX A

PROOF OF THEOREM 1

Recall that the weighted objective is defined as

$$J_\lambda(\pi_\theta) = \sum_{k=1}^K \lambda_k J_k(\pi_\theta),$$

where

$$J_k(\pi_\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, P_k)} [R(\tau)].$$

Since $\lambda \in \Delta_K$ is fixed and independent of θ , we can differentiate term-wise. Under standard regularity conditions that justify exchanging the gradient and expectation (e.g., bounded rewards and differentiable policy with finite variance), we obtain

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^K \lambda_k \nabla_\theta J_k(\pi_\theta).$$

For each environment k , applying the standard policy gradient theorem yields

$$\nabla_\theta J_k(\pi_\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, P_k)} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t | s_t) G_t^{(k)} \right],$$

where $G_t^{(k)}$ denotes the return-to-go under environment k . Substituting this expression into the weighted gradient yields

$$\nabla_\theta J_\lambda(\pi_\theta) = \sum_{k=1}^K \lambda_k \mathbb{E}_{\tau \sim (\pi_\theta, P_k)} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t | s_t) G_t^{(k)} \right].$$

This completes the proof.

APPENDIX B

PROOF OF THEOREM 2

We prove the result by analyzing the primal and dual updates separately.

A. Proof of primal convergence

Let $\mathcal{H}_m$ denote the filtration containing all randomness up to the beginning of iteration m , i.e.,

$$\mathcal{H}_m = \sigma(\theta^{(0)}, \lambda^{(0)}, g^{(0)}, \hat{g}_\lambda^{(0)}, \dots, g^{(m-1)}, \hat{g}_\lambda^{(m-1)}).$$

In particular, $\theta^{(m)}$ and $\lambda^{(m)}$ are $\mathcal{H}_m$ -measurable. Define

$$F_m := F(\theta^{(m)}, \lambda^{(m)}).$$

At iteration m , the policy update is given by

$$\theta^{(m+1)} = \theta^{(m)} + \eta_\pi g^{(m)}, \quad (32)$$

where $g^{(m)}$ is an unbiased estimator of $\nabla_\theta F(\theta^{(m)}, \lambda^{(m)})$ with bounded variance, as stated in Assumption 2. Since $F(\theta, \lambda)$ is L -smooth in θ by Assumption 1, we have

$$\begin{aligned} & F(\theta^{(m+1)}, \lambda^{(m)}) \\ & \geq F_m + \eta_\pi \langle \nabla_\theta F_m, g^{(m)} \rangle - \frac{L\eta_\pi^2}{2} \|g^{(m)}\|^2. \end{aligned}$$

Taking conditional expectation yields

$$\begin{aligned} & \mathbb{E} \left[F(\theta^{(m+1)}, \lambda^{(m)}) \mid \mathcal{H}_m \right] \\ & \geq F_m + \eta_\pi \|\nabla_\theta F_m\|^2 - \frac{L\eta_\pi^2}{2} \mathbb{E} \left[\|g^{(m)}\|^2 \mid \mathcal{H}_m \right]. \end{aligned}$$

Using variance decomposition together with the bounded variance assumption in Assumption 2, we obtain

$$\begin{aligned} \mathbb{E} \left[\|g^{(m)}\|^2 \mid \mathcal{H}_m \right] &= \|\nabla_\theta F_m\|^2 + \mathbb{E} \left[\|g^{(m)} - \nabla_\theta F_m\|^2 \mid \mathcal{H}_m \right] \\ &\leq \|\nabla_\theta F_m\|^2 + \sigma^2. \end{aligned}$$

Thus, we obtain

$$\begin{aligned} \mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)})] &\geq \mathbb{E}[F_m] + \eta_\pi \mathbb{E}\|\nabla_\theta F_m\|^2 \\ &\quad - \frac{L\eta_\pi^2}{2} \mathbb{E}\|\nabla_\theta F_m\|^2 - \frac{L\eta_\pi^2}{2} \sigma^2. \end{aligned}$$

Choosing $\eta_\pi \leq \frac{1}{L}$ yields

$$\mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)})] \geq \mathbb{E}[F_m] + \frac{\eta_\pi}{2} \mathbb{E}\|\nabla_\theta F_m\|^2 - C_1 \eta_\pi^2,$$

for some constant $C_1 > 0$. Rearranging gives

$$\mathbb{E}\|\nabla_\theta F_m\|^2 \leq \frac{2}{\eta_\pi} \mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)}) - F_m] + 2C_1 \eta_\pi.$$

Summing over $m = 0, \dots, M-1$ yields

$$\begin{aligned} \sum_{m=0}^{M-1} \mathbb{E}[\|\nabla_\theta F_m\|^2] &\leq \frac{2}{\eta_\pi} \sum_{m=0}^{M-1} \mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)}) - F_m] \\ &\quad + 2C_1 M \eta_\pi. \end{aligned} \quad (33)$$

By Lemma 1 below, we have

$$\sum_{m=0}^{M-1} \mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)}) - F_m] \leq C_2 + C_3 M \eta_\lambda,$$

and thus,

$$\frac{1}{M} \sum_{m=0}^{M-1} \mathbb{E}[\|\nabla_\theta F_m\|^2] \leq \mathcal{O}\left(\frac{1}{M\eta_\pi} + \frac{\eta_\lambda}{\eta_\pi} + \eta_\pi\right). \quad (34)$$

Choosing $\eta_\pi = \mathcal{O}(M^{-1/3})$ and $\eta_\lambda = \mathcal{O}(M^{-2/3})$ yields

$$\frac{1}{M} \sum_{m=0}^{M-1} \mathbb{E}[\|\nabla_\theta F_m\|^2] = \mathcal{O}(M^{-1/3}). \quad (35)$$

This proves the primal convergence claim.

Lemma 1: Under the projected gradient update for λ , the cumulative objective variation satisfies

$$\sum_{m=0}^{M-1} \mathbb{E}[F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m)}, \lambda^{(m)})] \leq C_2 + C_3 M \eta_\lambda,$$

for some constants $C_2, C_3 > 0$.

Proof: For each m , we decompose

$$\begin{aligned} &F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m)}, \lambda^{(m)}) \\ &= [F(\theta^{(m+1)}, \lambda^{(m+1)}) - F(\theta^{(m)}, \lambda^{(m)})] \\ &\quad + [F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m+1)}, \lambda^{(m+1)})]. \end{aligned}$$

Summing over $m = 0, \dots, M-1$, the first term telescopes:

$$\begin{aligned} &\sum_{m=0}^{M-1} [F(\theta^{(m+1)}, \lambda^{(m+1)}) - F(\theta^{(m)}, \lambda^{(m)})] \\ &= F(\theta^{(M)}, \lambda^{(M)}) - F(\theta^{(0)}, \lambda^{(0)}). \end{aligned}$$

Since the iterates $\{(\theta^{(m)}, \lambda^{(m)})\}$ remain in a bounded domain by Assumption 4, and $F(\theta, \lambda)$ is continuous, the above difference is bounded by a constant C_2 .

For the second term, note that $F(\theta, \lambda) = \sum_{k=1}^K \lambda_k J_k(\pi_\theta)$ is linear in λ . Thus, we have

$$\begin{aligned} &|F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m+1)}, \lambda^{(m+1)})| \\ &\leq \sum_{k=1}^K |J_k(\pi_{\theta^{(m+1)}})| \cdot |\lambda_k^{(m)} - \lambda_k^{(m+1)}|. \end{aligned}$$

Since rewards are bounded and $\gamma < 1$, the returns are uniformly bounded, i.e., $|J_k(\pi_\theta)| \leq J_{\max}$ for all k and θ . Hence, we obtain the upper bound

$$\begin{aligned} &|F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m+1)}, \lambda^{(m+1)})| \\ &\leq J_{\max} \|\lambda^{(m)} - \lambda^{(m+1)}\|_1. \end{aligned}$$

Under the projected gradient update for λ and using Assumption 3 (bounded variance of the return estimator), the dual gradient estimator is bounded, which implies

$$\|\lambda^{(m+1)} - \lambda^{(m)}\|_1 \leq C_\lambda \eta_\lambda$$

for some constant $C_\lambda > 0$. Therefore, we can get

$$\sum_{m=0}^{M-1} |F(\theta^{(m+1)}, \lambda^{(m)}) - F(\theta^{(m+1)}, \lambda^{(m+1)})| \leq J_{\max} C_\lambda M \eta_\lambda.$$

Combining the two parts and taking expectations yields the result with $C_3 = J_{\max} C_\lambda$. $\blacksquare$

B. Proof of dual no-regret property

The dual update is given by projected stochastic gradient descent:

$$\lambda^{(m+1)} = \Pi_{\Delta_K} \left(\lambda^{(m)} - \eta_\lambda \hat{g}_\lambda^{(m)} \right), \quad (36)$$

where $\hat{g}_\lambda^{(m)} = [J_1^{(m)}, \dots, J_K^{(m)}]^T$. The projection operator Π_{Δ_K} is non-expansive, i.e.,

$$\|\Pi_{\Delta_K}(x) - \Pi_{\Delta_K}(y)\| \leq \|x - y\|, \quad \forall x, y. \quad (37)$$

For each round m , we define

$$f_m(\lambda) := F(\theta^{(m)}, \lambda) = \sum_{k=1}^K \lambda_k J_k(\pi_{\theta^{(m)}}). \quad (38)$$

Conditioned on the history $\mathcal{H}_m$ prior to the dual update at round m , the parameter $\theta^{(m)}$ is fixed, and hence $f_m(\lambda)$ is a convex function of λ . Since f_m is linear in λ , it is convex over the simplex Δ_K . This corresponds to an adaptive (predictable) sequence of loss functions in online convex optimization [8], where each f_m depends on past iterates but remains fixed during the update.

Since the dual gradient estimator is given by $\hat{g}_\lambda^{(m)} = [J_1^{(m)}, \dots, J_K^{(m)}]^T$, it is formed by stacking the return estimators across environments. Under Assumption 3, the dual gradient estimator satisfies

$$\mathbb{E}[\hat{g}_\lambda^{(m)} \mid \mathcal{H}_m] = \nabla f_m(\lambda^{(m)}), \quad (39)$$

$$\mathbb{E}[\|\hat{g}_\lambda^{(m)}\|^2 \mid \mathcal{H}_m] \leq G^2, \quad (40)$$

for some constant $G > 0$, where the second inequality follows from the boundedness of returns together with bounded rewards and $\gamma < 1$.

By the standard analysis of projected online gradient descent [8], [9], for any fixed comparator $\lambda \in \Delta_K$, we have

$$\mathbb{E} \left[\sum_{m=0}^{M-1} \left(f_m(\lambda^{(m)}) - f_m(\lambda) \right) \right] \leq \frac{\|\lambda - \lambda^{(0)}\|^2}{2\eta_\lambda} + \frac{\eta_\lambda}{2} \sum_{m=0}^{M-1} \mathbb{E} \|\hat{g}_\lambda^{(m)}\|^2. \quad (41)$$

Since Δ_K is compact, there exists $D > 0$ such that $\|\lambda - \lambda^{(0)}\| \leq D$ for all $\lambda \in \Delta_K$. Therefore, we obtain

$$\mathbb{E} \left[\sum_{m=0}^{M-1} \left(f_m(\lambda^{(m)}) - f_m(\lambda) \right) \right] \leq \frac{D^2}{2\eta_\lambda} + \frac{\eta_\lambda G^2}{2} M. \quad (42)$$

Choosing $\eta_\lambda = \mathcal{O}(M^{-2/3})$ yields

$$\mathbb{E} \left[\sum_{m=0}^{M-1} \left(f_m(\lambda^{(m)}) - f_m(\lambda) \right) \right] = \mathcal{O}(M^{2/3}). \quad (43)$$

Since the bound holds for any fixed comparator $\lambda \in \Delta_K$, it also holds for the best fixed comparator in hindsight. Substituting $f_m(\lambda) = F(\theta^{(m)}, \lambda)$, we obtain

$$\mathbb{E} \left[\sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda^{(m)}) - \min_{\lambda \in \Delta_K} \sum_{m=0}^{M-1} F(\theta^{(m)}, \lambda) \right] = \mathcal{O}(M^{2/3}).$$

This establishes a sublinear regret bound for the dual variable.

APPENDIX C PROOF OF PROPOSITION 1

Let $\xi_k := g_k - \nabla_{\theta} J_k(\pi_{\theta})$. Under Assumption 2, which ensures conditional unbiasedness and bounded variance, we have

$$\mathbb{E}[\xi_k \mid \mathcal{H}_m] = 0, \quad \mathbb{E}[\|\xi_k\|^2 \mid \mathcal{H}_m] \leq \sigma^2.$$

Then, we obtain

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{k=1}^K \lambda_k \xi_k \right\|^2 \mid \mathcal{H}_m \right] &= \mathbb{E} \left[\left\langle \sum_{i=1}^K \lambda_i \xi_i, \sum_{j=1}^K \lambda_j \xi_j \right\rangle \mid \mathcal{H}_m \right] \\ &= \sum_{k=1}^K \lambda_k^2 \mathbb{E}[\|\xi_k\|^2 \mid \mathcal{H}_m] + \sum_{i \neq j} \lambda_i \lambda_j \mathbb{E}[\langle \xi_i, \xi_j \rangle \mid \mathcal{H}_m]. \end{aligned} \quad (44)$$

Under the conditional independence assumption on $\{\xi_k\}_{k=1}^K$ given $\mathcal{H}_m$, the cross terms vanish:

$$\mathbb{E}[\langle \xi_i, \xi_j \rangle \mid \mathcal{H}_m] = 0, \quad i \neq j.$$

Finally, we can get

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{k=1}^K \lambda_k \xi_k \right\|^2 \mid \mathcal{H}_m \right] &= \sum_{k=1}^K \lambda_k^2 \mathbb{E}[\|\xi_k\|^2 \mid \mathcal{H}_m] \\ &\leq \sigma^2 \sum_{k=1}^K \lambda_k^2. \end{aligned} \quad (45)$$

This completes the proof.

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